

Project Proposal Report: Web-Based Learning Management System



Course Code: ICT 3183

Organization: Rajasinghe Central College, Ruwanwella

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Table of Content

- 1. Introduction to Client**
- 2. Introduction to Project**
- 3. Problem Statement**
- 4. Aim & Objectives**
- 5. Scope of Work**
- 6. Methodology & Technology Stack**
- 7. Project Timeline**
- 8. Project Timeline**
- 9. Estimated Budget**
- 10. Risk Assessment**
- 11. Project Team**
- 12. Hardware & Software Requirements**
- 13. Conclusion**

1. Introduction to Client

Rajasinghe Central College in Ruwanwella is a leading National School dedicated to secondary education (Grades 6–13). The institution serves as a hub for academic excellence, admitting high-achieving students who qualify through the Grade 5 Scholarship Examination.

The college provides a comprehensive curriculum, including all major Advanced Level streams: Math's, Science, Commerce, Arts, and Technology. By balancing rigorous academics with sports and leadership programs, Rajasinghe Central continues to be a premier seat of learning in the region.

2. Introduction to Project

- This project proposes a web-based Learning Management System (LMS) integrated with the official school website.
- The system is developed for Rajasinghe Central College.
- It will support digital teaching, learning, and communication.
- Teachers, students, and administrators will use one centralized platform.

3. Problem Statement

- Learning materials are shared manually.
- No centralized system for assignments and notices.
- Student progress is not tracked digitally.
- The school does not have a modern official website.
- Communication between teachers and students is slow.

4. Aim & Objectives

Aim-

To develop a comprehensive web-based Learning Management System (LMS) seamlessly integrated with the official school website for Rajasinghe Central College, fostering a dynamic and accessible digital learning environment.

Objectives:

- Provide online access to learning materials.
- Enable digital assignment submission.
- Improve school-wide communication.
- Create an official school website.
- Provide a secure login system.
- Implement student record management.

5. Scope of Work

Included:	Excluded:
• School website integration.	• Mobile application.
• Student & teacher management.	• Online payment system.
• Learning materials management.	• SMS notification system.
• Assignment submission & grading.	
• Notices & announcements.	
• Provide Online Access to Learning Materials	

6. System Requirements

Functional Requirements:	Non-Functional Requirements:
• Role-based login.	• Responsive design.
• Learning material management.	• High performance.
• Assignment management.	• Data security.
• Announcements & notices.	• User-friendly UI.
• Student data tracking.	• Data integrity.
• Administrative controls.	• Scalability.
• Public website.	• Maintainability.

7. Methodology & Technology Stack

Methodology:

- Agile

Technologies:

- React.js
- Tailwind CSS
- Node.js & Express.js
- MongoDB

Tools:

- Postman
- Jira
- GitHub
- Figma

8. Project Timeline

- Requirement Analysis – 1 Week
- Design – 2 Weeks
- Development – 8 Weeks
- Testing – 2 Weeks
- Deployment – 1 Week

9. Estimated Budget

- Domain: 5,000
- Hosting: 10,000
- Miscellaneous: 8,000

Total: 23,000

10. Risk Assessment

- Internet issues – Mitigation via planning.
- Requirement changes – Agile control.
- Time issues – Monitoring.
- Data security – Security & backups.

11. Project Team

- P.C.P Premathilaka – Project Lead & Backend
- W.A.V.M Siriwardhana – UI/UX & Frontend
- B.N.S Lakmal – QA & Tester
- G.H.W.K Kumara – DBA & DevOps

12. Hardware & Software Requirements

For Development Team

Hardware

- Desktop or Laptop (Minimum 8GB RAM recommended)
- Reliable Internet connection

Software

- Visual Studio Code
- Node.js
- MongoDB
- Web Browser (Chrome / Edge)
- GitHub Desktop or CLI
- React.js development environment
- Tailwind CSS configuration

For Rajasinghe Central College (Client)

Hardware

- Desktop / Laptop
- Smartphone (optional, for mobile access)
- Stable Internet connection

Software

- Modern Web Browser (Chrome, Edge, Firefox, Safari)
- Internet connection

13. Conclusion:

This project is poised to modernize education at Rajasinghe Central College by creating a digitally advanced learning hub that enhances engagement and provides a robust digital foundation for future growth